



Figure 9.6.1. The Boeing 737 short-range transport, showing sweep and dihedral. (Photograph by Barry Santana.)

9.6 The Numerical Lifting Lifting-line Method

Prandtl's classical lifting-line equation, expressed in Eq. (9.1.24), applies only to a single lifting surface with no sweep and no dihedral. However, the wing and tail of most conventional aircraft usually have at least some degree of sweep and dihedral, as is seen in the Boeing 737 shown in Fig 9.6.1. A wing is said to have positive sweep if the wing slopes back from the root to the tip, with the wingtips being aft of the root. A wing is said to have positive dihedral if the wing slopes up from the root to the tip, with the wingtips being above the root. Negative dihedral is usually called anhedral. Because a wing can also have taper and/or geometric twist, defining the sweep and dihedral angles can be somewhat ambiguous. For a conventional tapered wing, the sweep angle for the leading edge is greater than that for the trailing edge. For a wing with positive geometric twist, called washout, the dihedral angle for the trailing edge is greater than that for the leading edge. Furthermore, neither the sweep nor the dihedral must remain constant over the semi-span of a lifting surface. For this reason, local sweep and dihedral angles are defined in terms of the wing quarter-chord line and the orientation of the local airfoil section, as shown in Fig. 9.6.2.

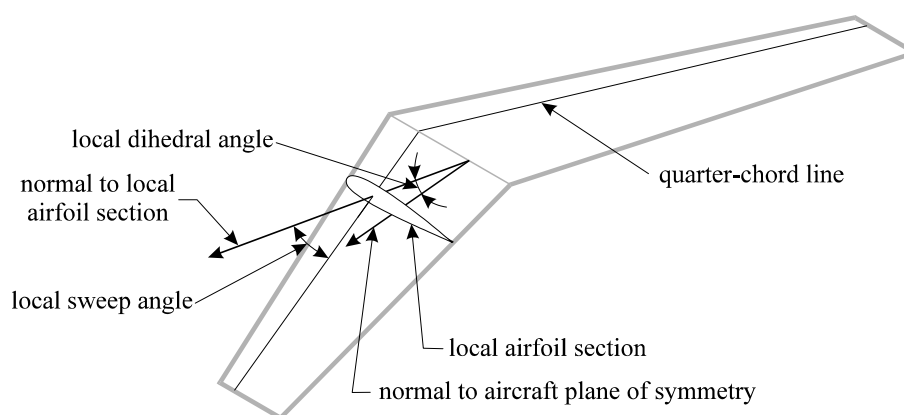


Figure 9.6.2. The local sweep and dihedral angles.

When constant dihedral is added to a wing, each side of the wing is rotated about the root as a solid body. The rotation of the left-hand side is opposite to that of the right-hand side so that both wingtips are raised above the root and brought closer together. Since dihedral is a solid body rotation, the local airfoil sections rotate with the quarter-chord line. For a wing with no dihedral, a vector drawn normal to a local airfoil section is also normal to the aircraft plane of symmetry. As dihedral is added to the wing, both the quarter-chord and the airfoil section-normal are rotated upward, so that the normal to the local airfoil section always forms an angle equal to the dihedral angle with the normal to the aircraft plane of symmetry. For example, the vertical stabilizer of a conventional airplane usually has a dihedral angle of 90 degrees. For the rarely encountered case of an aircraft that does not have a plane of symmetry, the dihedral angle must be defined relative to some other defined reference plane.

Unlike dihedral, constant sweep is not a solid body rotation of each semi-span. For a wing with no sweep the normal to each local airfoil section is aligned with the wing quarter-chord line and the quarter-chord line is straight over each semi-span of the wing. As constant positive sweep is added to the wing, the quarter-chord line for each semi-span is rotated back, moving the wingtips aft of the root. However, the local airfoil sections do not rotate with the quarter chord line. Each airfoil section is simply translated straight back to the new quarter-chord location with the orientation of the section remaining parallel to its original orientation. To maintain constant planform area as sweep is added to a wing, the length of the quarter-chord line is increased, so that the distance from wingtip to wingtip does not change. The sweep angle is defined to be the angle that the normal to the airfoil section makes with the quarter-chord line. Thus, the sweep angle for a wing, with no dihedral, is measured relative to a horizontal line while the sweep angle for a vertical stabilizer, with 90 degrees dihedral, is measured relative to a vertical line.

Most airplanes are designed with some sweep and/or dihedral in the wing and/or the horizontal stabilizer. While the reasons for using sweep and dihedral will be left for a later discussion, it is clear from looking at the myriad of airplanes parked at any airport that some means for analyzing the effects of sweep and dihedral is needed. In addition, the downwash created by one lifting surface, such as the wing, has a dramatic effect on the performance of other lifting surfaces, such as the tail. Thus, some means for predicting the effects of this interaction is also needed.

In the development of Prandtl's lifting-line equation, a single lifting surface with no sweep or dihedral was assumed. The lifting-line was confined to the z -axis and the trailing vortex sheet was assumed to be in the x - z plane. This significantly simplified the expressions for downwash and induced angle of attack. In order to use lifting-line theory to study the effects of sweep, dihedral, and/or the interactions between lifting surfaces, the theory must be generalized to allow for an arbitrary position and orientation of both the lifting-line and the trailing vortex sheet. In this section we shall present a numerical lifting-line method developed by Phillips and Snyder (2000) that can be used to obtain a potential flow solution for the forces and moments acting on a system of lifting surfaces, each with arbitrary position and orientation.

Since the vortex filaments used to synthesize finite wings cannot be straight over their entire length, these vortex filaments are often broken up into finite segments. Consider first an arbitrary straight segment of a vortex filament that extends from point-1 to point-2, as shown in Fig. 9.6.3. We wish to determine the velocity vector induced at point-3 by this straight vortex segment. Let point-4 be an arbitrary point on the vortex segment, between point-1 and point-2. Let ζ be the distance from point-1 to point-4 divided by the distance from point-1 to point-2. Let $d\vec{V}$ be the differential velocity vector induced at point-3 by the directed differential segment of the filament, $d\vec{l}$, located at point-4. From the Biot-Savart law, Eq. (9.1.1), we have

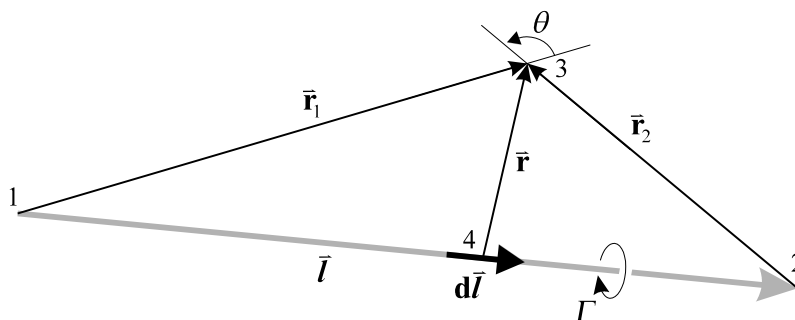


Figure 9.6.3. Geometric relationships for the velocity induced by a straight vortex segment.

$$\mathbf{d}\bar{\mathbf{V}} = \frac{\Gamma \mathbf{d}\bar{\mathbf{l}} \times \bar{\mathbf{r}}}{4\pi |\bar{\mathbf{r}}|^3} \quad (9.6.1)$$

where Γ is the vortex filament strength and $\bar{\mathbf{r}}$ is the spatial vector from point-4 to point-3.

If $\bar{\mathbf{l}}$ is the spatial vector from point-1 to point-2, $\bar{\mathbf{r}}_1$ is the spatial vector from point-1 to point-3, and $\bar{\mathbf{r}}_2$ is the spatial vector from point-2 to point-3, then from Fig. 9.6.3

$$\mathbf{d}\bar{\mathbf{l}} = \bar{\mathbf{l}} d\zeta \quad (9.6.2)$$

and

$$\bar{\mathbf{r}} = \bar{\mathbf{r}}_1 - \zeta \bar{\mathbf{l}} \quad (9.6.3)$$

From Eq. (9.6.2) and (9.6.3) we can write

$$\mathbf{d}\bar{\mathbf{l}} \times \bar{\mathbf{r}} = (\bar{\mathbf{l}} d\zeta) \times (\bar{\mathbf{r}}_1 - \zeta \bar{\mathbf{l}}) = (\bar{\mathbf{l}} \times \bar{\mathbf{r}}_1 - \zeta \bar{\mathbf{l}} \times \bar{\mathbf{l}}) d\zeta \quad (9.6.4)$$

Since the cross product of any vector with itself is zero, Eq. (9.6.4) becomes

$$\mathbf{d}\bar{\mathbf{l}} \times \bar{\mathbf{r}} = \bar{\mathbf{l}} \times \bar{\mathbf{r}}_1 d\zeta \quad (9.6.5)$$

From Eq. (9.6.3) we have

$$\begin{aligned} |\bar{\mathbf{r}}|^3 &= \left[(r_{1x} - \zeta l_x)^2 + (r_{1y} - \zeta l_y)^2 + (r_{1z} - \zeta l_z)^2 \right]^{3/2} \\ &= \left[(r_{1x}^2 + r_{1y}^2 + r_{1z}^2) - 2(l_x r_{1x} + l_y r_{1y} + l_z r_{1z})\zeta + (l_x^2 + l_y^2 + l_z^2)\zeta^2 \right]^{3/2} \\ &= (\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_1 - 2\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1 \zeta + \bar{\mathbf{l}} \cdot \bar{\mathbf{l}} \zeta^2)^{3/2} = (r_1^2 - 2\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1 \zeta + l^2 \zeta^2)^{3/2} \end{aligned} \quad (9.6.6)$$

Using Eqs. (9.6.5) and (9.6.6) in Eq. (9.6.1), the differential velocity vector induced at point-3 by a differential segment of the filament located at point-4 is

$$\mathbf{d}\bar{\mathbf{V}} = \frac{\Gamma \bar{\mathbf{l}} \times \bar{\mathbf{r}}_1 d\zeta}{4\pi (r_1^2 - 2\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1 \zeta + l^2 \zeta^2)^{3/2}} \quad (9.6.7)$$

The velocity vector induced at point-3 by the complete vortex segment extending from point-1 to point-2 is then

$$\bar{\mathbf{V}} = \frac{\Gamma}{4\pi} \bar{\mathbf{l}} \times \bar{\mathbf{r}}_1 \int_{\zeta=0}^1 \frac{d\zeta}{(r_1^2 - 2\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1 \zeta + l^2 \zeta^2)^{3/2}} \quad (9.6.8)$$

The integration with respect to ζ is easily carried out to yield

$$\begin{aligned} \bar{\mathbf{V}} &= \frac{\Gamma (\bar{\mathbf{l}} \times \bar{\mathbf{r}}_1)}{4\pi [l^2 r_1^2 - (\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1)^2]} \left[\frac{l^2 \zeta - \bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1}{(r_1^2 - 2\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1 \zeta + l^2 \zeta^2)^{1/2}} \right]_{\zeta=0}^1 \\ &= \frac{\Gamma (\bar{\mathbf{l}} \times \bar{\mathbf{r}}_1)}{4\pi [l^2 r_1^2 - (\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1)^2]} \left[\frac{l^2 - \bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1}{(r_1^2 - 2\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1 + l^2)^{1/2}} - \frac{-\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1}{(r_1^2)^{1/2}} \right] \\ &= \frac{\Gamma (\bar{\mathbf{l}} \times \bar{\mathbf{r}}_1)}{4\pi [l^2 r_1^2 - (\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1)^2]} \left[\frac{-\bar{\mathbf{l}} \cdot (\bar{\mathbf{r}}_1 - \bar{\mathbf{l}})}{[(\bar{\mathbf{r}}_1 - \bar{\mathbf{l}}) \cdot (\bar{\mathbf{r}}_1 - \bar{\mathbf{l}})]^{1/2}} + \frac{\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1}{r_1} \right] \end{aligned} \quad (9.6.9)$$

From Fig. 9.6.3 we can see that $\bar{\mathbf{r}}_1$ is the vector sum of $\bar{\mathbf{l}}$ and $\bar{\mathbf{r}}_2$. Thus, we have

$$\bar{\mathbf{l}} = \bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2 \quad (9.6.10)$$

$$\begin{aligned} l^2 r_1^2 - (\bar{\mathbf{l}} \cdot \bar{\mathbf{r}}_1)^2 &= (\bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2) \cdot (\bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2) r_1^2 - [(\bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2) \cdot \bar{\mathbf{r}}_1]^2 \\ &= (\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2 - \bar{\mathbf{r}}_2 \cdot \bar{\mathbf{r}}_1 + \bar{\mathbf{r}}_2 \cdot \bar{\mathbf{r}}_2) r_1^2 - (\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2 \cdot \bar{\mathbf{r}}_1)^2 \\ &= (r_1^2 - 2\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2 + r_2^2) r_1^2 - (r_1^2 - \bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2)^2 = r_1^2 r_2^2 - (\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2)^2 \end{aligned} \quad (9.6.11)$$

and

$$\bar{\mathbf{l}} \times \bar{\mathbf{r}}_1 = (\bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2) \times \bar{\mathbf{r}}_1 = \bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2 \times \bar{\mathbf{r}}_1 = -\bar{\mathbf{r}}_2 \times \bar{\mathbf{r}}_1 = \bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2 \quad (9.6.12)$$

Using Eqs. (9.6.10), (9.6.11), and (9.6.12) in Eq. (9.6.9), the velocity vector induced at an arbitrary point in space, by a straight vortex segment, is given by

$$\bar{\mathbf{V}} = \frac{\Gamma}{4\pi} \frac{\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2}{r_1^2 r_2^2 - (\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2)^2} \bar{\mathbf{l}} \cdot \left(\frac{\bar{\mathbf{r}}_1}{r_1} - \frac{\bar{\mathbf{r}}_2}{r_2} \right) \quad (9.6.13)$$

By making use of certain trigonometric identities, Eq. (9.6.13) can be written in several different forms. The dot product of any two vectors is the product of the vector magnitudes multiplied by the cosine of the angle between the vectors. Also the magnitude of the cross product of two vectors is the product of vector magnitudes multiplied by the sine of the angle between the vectors. Thus, we can write

$$r_1^2 r_2^2 - (\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2)^2 = r_1^2 r_2^2 (1 - \cos^2 \theta) = r_1^2 r_2^2 \sin^2 \theta = |\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2|^2 \quad (9.6.14)$$

where θ is the angle from $\bar{\mathbf{r}}_1$ to $\bar{\mathbf{r}}_2$, as shown in Fig. 9.6.3. Using Eq. (9.6.14) in Eq. (9.6.13), the velocity induced by the vortex segment can also be written

$$\bar{\mathbf{V}} = \frac{\Gamma}{4\pi} \frac{\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2}{|\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2|^2} \bar{\mathbf{l}} \cdot \left(\frac{\bar{\mathbf{r}}_1}{r_1} - \frac{\bar{\mathbf{r}}_2}{r_2} \right) \quad (9.6.15)$$

This is the form most commonly found in modern textbooks. However, neither the form given by Eq. (9.6.13) nor that given by Eq. (9.6.15) is particularly useful for numerical calculations. In both forms, the induced velocity is indeterminate whenever $\bar{\mathbf{r}}_1$ and $\bar{\mathbf{r}}_2$ are collinear, even if the point lies outside the vortex segment. To eliminate this division by zero for points that are not on the vortex segment, we can use Eqs. (9.6.10) and (9.6.14) to obtain

$$\begin{aligned} \frac{\bar{\mathbf{l}}}{|\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2|^2} \cdot \left(\frac{\bar{\mathbf{r}}_1}{r_1} - \frac{\bar{\mathbf{r}}_2}{r_2} \right) &= \frac{\bar{\mathbf{r}}_1 - \bar{\mathbf{r}}_2}{|\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2|^2} \cdot \left(\frac{\bar{\mathbf{r}}_1}{r_1} - \frac{\bar{\mathbf{r}}_2}{r_2} \right) = \frac{1}{r_1^2 r_2^2 \sin^2 \theta} \left(r_1 - \frac{\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2}{r_2} - \frac{\bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2}{r_1} + r_2 \right) \\ &= \frac{(r_1 + r_2)(1 - \cos \theta)}{r_1^2 r_2^2 (1 - \cos^2 \theta)} = \frac{r_1 + r_2}{r_1^2 r_2^2 (1 + \cos \theta)} = \frac{r_1 + r_2}{r_1 r_2 (r_1 r_2 + \bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2)} \end{aligned} \quad (9.6.16)$$

Using Eq. (9.6.16) in Eq. (9.6.15), the **velocity vector induced at an arbitrary point in space, by any straight vortex segment**, is

$$\bar{\mathbf{V}} = \frac{\Gamma}{4\pi} \frac{(r_1 + r_2)(\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2)}{r_1 r_2 (r_1 r_2 + \bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2)} \quad (9.6.17)$$

where Γ is the vortex strength, $\bar{\mathbf{r}}_1$ is the spatial vector from the beginning of the segment to the arbitrary point in space, and $\bar{\mathbf{r}}_2$ is the spatial vector from the end of the segment to the arbitrary point in space. Notice that, unlike Eqs. (9.6.13) and (9.6.15), the induced velocity computed from Eq. (9.6.17) is not indeterminate when the angle from $\bar{\mathbf{r}}_1$ to $\bar{\mathbf{r}}_2$ is zero. It is, however, still singular when this angle is $\pm\pi$.

Equation (9.6.17) is the fundamental equation that allows us to calculate the velocity induced by a general vortex filament. The bound vortex filaments associated with any lifting surface can be approximated as a series of these straight vortex segments. The trailing vortex sheet can also be approximated with straight vortex segments. However, a trailing vortex sheet is often synthesized from semi-infinite vortex filaments.

The velocity that is induced by a straight semi-infinite vortex segment can be determined as a special case of Eq. (9.6.17), by taking the limit as r_2 approaches infinity. To facilitate taking this limit, Eq. (9.6.17) can be rearranged as

$$\bar{\mathbf{V}} = \frac{\Gamma}{4\pi} \frac{\left(\frac{r_1}{r_2} + 1\right) \left(\bar{\mathbf{r}}_1 \times \frac{\bar{\mathbf{r}}_2}{r_2}\right)}{r_1 \left(r_1 + \bar{\mathbf{r}}_1 \cdot \frac{\bar{\mathbf{r}}_2}{r_2}\right)} \quad (9.6.18)$$

In the limit as r_2 approaches infinity,

$$\frac{\bar{\mathbf{r}}_2}{r_2} \rightarrow -\bar{\mathbf{u}}_\infty \quad (9.6.19)$$

where $\bar{\mathbf{u}}_\infty$ is the unit vector in the direction of the vortex segment. Substituting Eq. (9.6.19) into Eq. (9.6.18) and taking the limit as r_2 approaches infinity, the **velocity vector induced at an arbitrary point in space by a straight semi-infinite vortex segment** is given by

$$\bar{\mathbf{V}} = \frac{\Gamma}{4\pi} \frac{\bar{\mathbf{u}}_\infty \times \bar{\mathbf{r}}_1}{r_1 (r_1 - \bar{\mathbf{u}}_\infty \cdot \bar{\mathbf{r}}_1)} \quad (9.6.20)$$

where Γ is the vortex strength, $\bar{\mathbf{u}}_\infty$ is the unit vector in the direction from the beginning of the segment toward the end at infinity, and $\bar{\mathbf{r}}_1$ is the spatial vector from the beginning of the segment to the arbitrary point in space.

A first order numerical lifting-line method can be obtained by synthesizing a system of lifting surfaces using a composite of horseshoe shaped vortices. The continuous distribution of bound vorticity over the surface of each wing, as well as the continuous distribution of free vorticity in the trailing vortex sheets, is approximated by a finite number of discrete horseshoe vortices, as shown in Fig. 9.6.4. The bound portion of each horseshoe vortex is placed coincident with the wing quarter-chord line and is, thus, aligned with the local sweep and dihedral. The trailing portion of each horseshoe vortex is aligned with the trailing vortex sheet. In obtaining the classical lifting-line

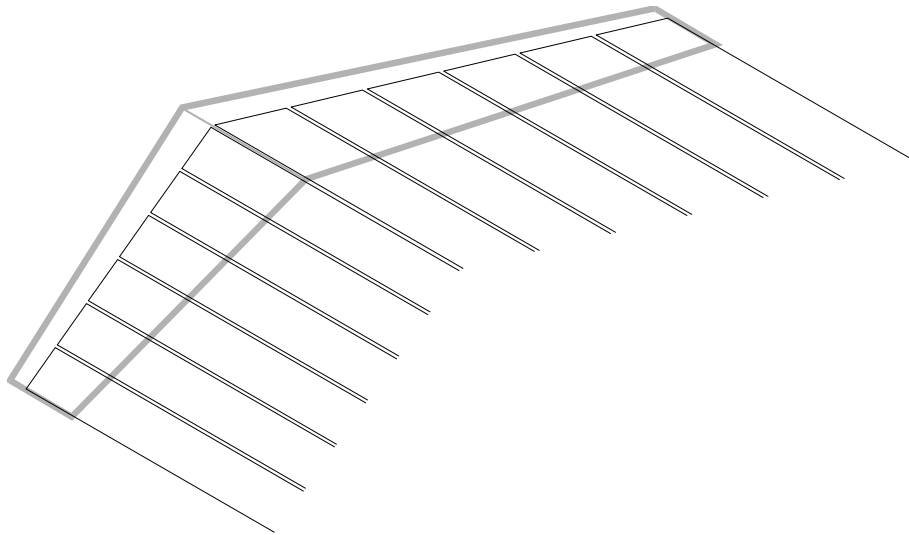


Figure 9.6.4. Horseshoe vortices distributed along the lifting-line of a finite wing with sweep and dihedral.

solution for a single lifting surface with no sweep or dihedral, Prandtl assumed the trailing vortex sheet to be aligned with the wing chord. This was done to facilitate obtaining an analytical solution. In obtaining a numerical solution, there is little advantage in aligning the trailing vortex sheet with a vehicle axis such as the chord line. More correctly, the trailing vortex sheet should be aligned with the freestream. This is easily done in the numerical solution by setting $\bar{\mathbf{u}}_\infty$ equal to the unit vector in the direction of the freestream. When using this method to predict the forces and moments on a single lifting surface, there is very little difference between the results obtained from slightly different orientations of the trailing vortex sheet.

In Fig. 9.6.4, a small gap is shown between the left-hand trailing segment of one horseshoe vortex and the right-hand trailing segment of the next. This is for display purpose only. In reality, the left-hand corner of one horseshoe and the right-hand corner of the next are both placed on the same nodal point. Thus, except at the wingtips, each trailing vortex segment is coincident with another trailing segment from the adjacent vortex. If two adjacent vortices have exactly the same strength, then the two coincident trailing segments exactly cancel, since one has clockwise rotation and the other has counterclockwise rotation. The net vorticity that is shed from the wing at any internal node is simply the difference in vorticity of the two adjacent vortices that share that node.

Using Eqs. (9.6.17) and (9.6.20), we can now calculate the velocity induced at an arbitrary point in space (x, y, z) by a general horseshoe vortex. As shown in Fig. 9.6.5, a general horseshoe vortex is completely defined by two nodal points, (x_1, y_1, z_1) and (x_2, y_2, z_2) , a trailing unit vector, $\bar{\mathbf{u}}_\infty$, and a vortex strength, Γ . The horseshoe vortex starts at the fluid boundary, an infinite distance downstream. The inbound trailing vortex segment is directed along the vector $-\bar{\mathbf{u}}_\infty$ to node-1 at (x_1, y_1, z_1) . The bound vortex segment is directed along the wing quarter-chord line from node-1 to node-2 at (x_2, y_2, z_2) . The outbound trailing vortex segment is directed along the vector $\bar{\mathbf{u}}_\infty$ from node-2 to the fluid boundary, an infinite distance back downstream.

The velocity induced by the entire horseshoe vortex is simply the vector sum of the velocities induced by each of the three linear segments that make up the horseshoe. In Eq. (9.6.20) the vorticity vector is assumed to point in the direction of $\bar{\mathbf{u}}_\infty$. However, the inbound trailing vortex segment can be treated like an outbound segment with negative circulation. Thus, from Eqs. (9.6.17) and (9.6.20), the **velocity vector induced at an arbitrary point in space, by a complete horseshoe vortex** is

$$\bar{\mathbf{V}} = \frac{\Gamma}{4\pi} \left[\frac{\bar{\mathbf{u}}_\infty \times \bar{\mathbf{r}}_2}{r_2(r_2 - \bar{\mathbf{u}}_\infty \cdot \bar{\mathbf{r}}_2)} + \frac{(r_1 + r_2)(\bar{\mathbf{r}}_1 \times \bar{\mathbf{r}}_2)}{r_1 r_2 (r_1 r_2 + \bar{\mathbf{r}}_1 \cdot \bar{\mathbf{r}}_2)} - \frac{\bar{\mathbf{u}}_\infty \times \bar{\mathbf{r}}_1}{r_1(r_1 - \bar{\mathbf{u}}_\infty \cdot \bar{\mathbf{r}}_1)} \right] \quad (9.6.21)$$

where $\bar{\mathbf{u}}_\infty$ is the unit vector in the direction of the freestream, $\bar{\mathbf{r}}_1$ is the spatial vector from (x_1, y_1, z_1) to (x, y, z) , and $\bar{\mathbf{r}}_2$ is the spatial vector from (x_2, y_2, z_2) to (x, y, z) ,

$$\bar{\mathbf{r}}_1 = (x - x_1)\bar{\mathbf{i}}_x + (y - y_1)\bar{\mathbf{i}}_y + (z - z_1)\bar{\mathbf{i}}_z \quad (9.6.22)$$

$$\bar{\mathbf{r}}_2 = (x - x_2)\bar{\mathbf{i}}_x + (y - y_2)\bar{\mathbf{i}}_y + (z - z_2)\bar{\mathbf{i}}_z \quad (9.6.23)$$

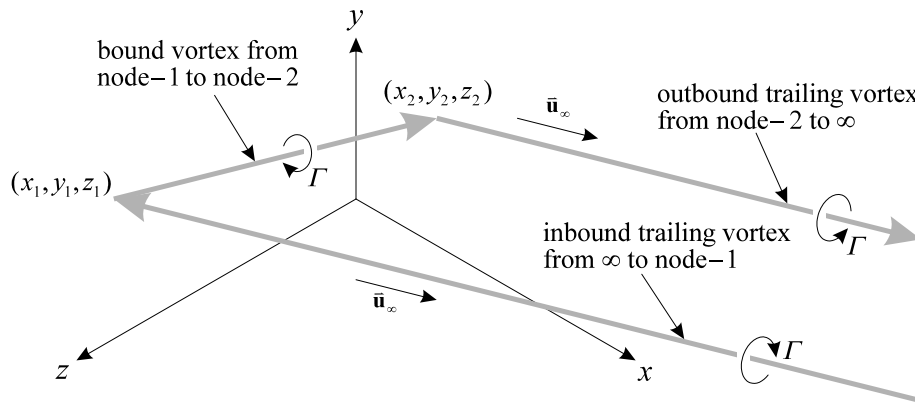


Figure 9.6.5. General horseshoe vortex.

When a system of lifting surfaces is synthesized using N of these horseshoe vortices, in a manner similar to that shown in Fig. 9.6.4, Eq. (9.6.21) can be used to determine the resultant velocity induced at any point in space, if the vortex strength of each horseshoe vortex is known. However, these strengths are not known a priori. To compute the strengths of the N vortices, we must have a system of N equations relating these N strengths to some known properties of the wing. For this we turn to the three-dimensional vortex lifting law. This vortex lifting law requires that, for any potential flow containing vortex filaments, the force per unit length exerted on the surroundings at any point along a vortex filament is equal to the product of the fluid density, the vortex strength, and the cross product of the local fluid velocity vector with the unit vector in the direction of local vorticity,

$$d\bar{\mathbf{F}} = \rho \Gamma \bar{\mathbf{V}} \times d\bar{\mathbf{l}} \quad (9.6.24)$$

where $d\bar{\mathbf{F}}$ is the differential aerodynamic force, ρ the fluid density, Γ is the vortex strength, $\bar{\mathbf{V}}$ is the local fluid velocity, and $d\bar{\mathbf{l}}$ is the directed differential vortex length vector. A proof of this three-dimensional vortex lifting law is given in App. F.

If flow over a finite wing is synthesized from a uniform flow combined with horseshoe vortices placed along the wing quarter-chord, from Eq. (9.6.21), the local velocity induced at a control point placed anywhere along the bound segment of horseshoe vortex i is

$$\bar{\mathbf{V}}_i = \bar{\mathbf{V}}_\infty + \sum_{j=1}^N \frac{\Gamma_j \bar{\mathbf{v}}_{ji}}{\bar{c}_j} \quad (9.6.25)$$

where $\bar{\mathbf{V}}_\infty$ is the velocity of the uniform flow, Γ_j is the strength of horseshoe vortex j , $\bar{\mathbf{v}}_{ji}$ is a dimensionless velocity that would be induced at control point i by horseshoe vortex j , having a unit strength,

$$\bar{\mathbf{v}}_{ji} = \begin{cases} \frac{\bar{c}_j}{4\pi} \left[\frac{\bar{\mathbf{u}}_\infty \times \bar{\mathbf{r}}_{j2i}}{r_{j2i}(r_{j2i} - \bar{\mathbf{u}}_\infty \cdot \bar{\mathbf{r}}_{j2i})} + \frac{(r_{ji} + r_{j2i})(\bar{\mathbf{r}}_{ji} \times \bar{\mathbf{r}}_{j2i})}{r_{ji}r_{j2i}(r_{ji}r_{j2i} + \bar{\mathbf{r}}_{ji} \cdot \bar{\mathbf{r}}_{j2i})} - \frac{\bar{\mathbf{u}}_\infty \times \bar{\mathbf{r}}_{ji}}{r_{ji}(r_{ji} - \bar{\mathbf{u}}_\infty \cdot \bar{\mathbf{r}}_{ji})} \right], & j \neq i \\ \frac{\bar{c}_j}{4\pi} \left[\frac{\bar{\mathbf{u}}_\infty \times \bar{\mathbf{r}}_{j2i}}{r_{j2i}(r_{j2i} - \bar{\mathbf{u}}_\infty \cdot \bar{\mathbf{r}}_{j2i})} - \frac{\bar{\mathbf{u}}_\infty \times \bar{\mathbf{r}}_{ji}}{r_{ji}(r_{ji} - \bar{\mathbf{u}}_\infty \cdot \bar{\mathbf{r}}_{ji})} \right], & j = i \end{cases} \quad (9.6.26)$$

$\bar{\mathbf{r}}_{ji}$ is the spatial vector from node-1 of horseshoe vortex j to the control point of horseshoe vortex i , $\bar{\mathbf{r}}_{j2i}$ is the spatial vector from node-2 of horseshoe vortex j to the control point of horseshoe vortex i , and $\bar{\mathbf{u}}_\infty$ is the unit vector in the direction of the freestream. At this point, $\bar{c}_j$ could be any characteristic length associated with the wing section aligned with horseshoe vortex j . This characteristic length is simply used to nondimensionalize Eq. (9.6.26) and has no effect on the induced velocity. The choice of $\bar{c}_j$ will be addressed later. The bound vortex segment is excluded from Eq. (9.6.26), when $j=i$, because a straight vortex segment induces no downwash along its own length.

From Eqs. (9.6.24) and (9.6.25), the aerodynamic force acting on a spanwise differential section of the wing located at control point i is given by

$$d\bar{\mathbf{F}}_i = \rho \Gamma_i \left(\bar{\mathbf{V}}_\infty + \sum_{j=1}^N \frac{\Gamma_j}{\bar{c}_j} \bar{\mathbf{v}}_{ji} \right) \times d\bar{\mathbf{l}}_i \quad (9.6.27)$$

Allowing for the possibility of flap deflection, the local section lift coefficient for the airfoil section located at control point i is a function of local angle of attack and local flap deflection,

$$\tilde{C}_{Li} = \tilde{C}_{Li}(\alpha_i, \delta_i) \quad (9.6.28)$$

where $\tilde{C}_{Li}$, α_i , and δ_i are respectively the local airfoil section lift coefficient, the local angle of attack, and the local flap deflection, all evaluated for the airfoil section aligned with control point i . Defining $\bar{\mathbf{u}}_{ni}$ and $\bar{\mathbf{u}}_{ai}$ to be the local unit normal and axial vectors for the airfoil section located at control point- i , as shown in Fig. 9.6.6, the local angle of attack at control point- i can be written as

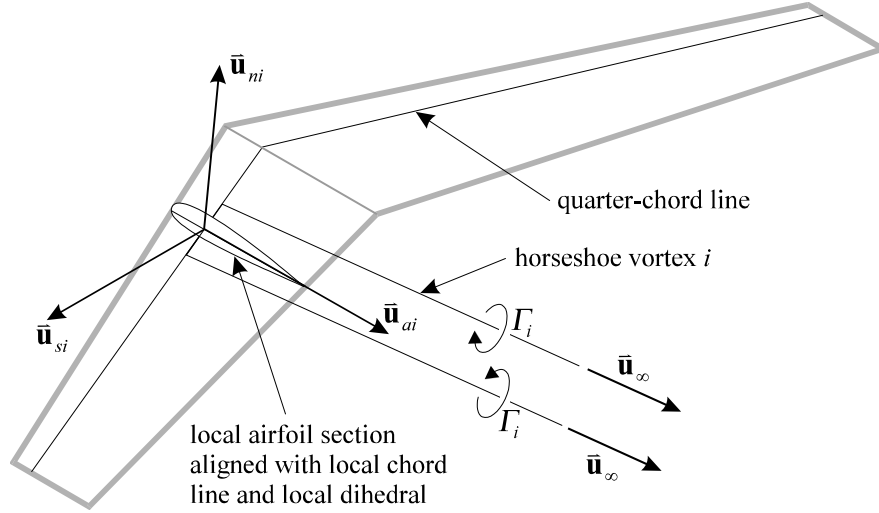


Figure 9.6.6. The unit vectors describing the orientation of the local airfoil section.

$$\alpha_i = \tan^{-1} \left(\frac{\bar{\mathbf{V}}_i \cdot \bar{\mathbf{u}}_{ni}}{\bar{\mathbf{V}}_i \cdot \bar{\mathbf{u}}_{ai}} \right) \quad (9.6.29)$$

If the relation implied by Eq. (9.6.28) is known at each section of the wing, the magnitude of the aerodynamic force acting on a spanwise differential section of the wing located at control point- i can be written as

$$|\mathbf{d}\bar{\mathbf{F}}_i| = \frac{1}{2} \rho V_\infty^2 \tilde{C}_{Li}(\alpha_i, \delta_i) dS_i \quad (9.6.30)$$

where dS_i is a spanwise differential planform area element located at control point- i . Setting the magnitude of the force obtained from Eq. (9.6.27) equal to that obtained from Eq. (9.6.30), applying Eq. (9.6.25) to Eq. (9.6.29), and rearranging, we can write

$$2 \left[\left(\bar{\mathbf{v}}_\infty + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} G_j \right) \times \bar{\boldsymbol{\zeta}}_i \right] \cdot G_i - \tilde{C}_{Li}(\alpha_i, \delta_i) = 0 \quad (9.6.31)$$

where

$$\bar{\mathbf{v}}_\infty \equiv \frac{\bar{\mathbf{V}}_\infty}{V_\infty}, \quad \bar{\boldsymbol{\zeta}}_i \equiv \bar{c}_i \frac{d\bar{\mathbf{l}}_i}{dS_i}, \quad G_i \equiv \frac{\Gamma_i}{\bar{c}_i V_\infty}$$

and

$$\alpha_i = \tan^{-1} \left[\frac{\left(\bar{\mathbf{v}}_\infty + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} G_j \right) \cdot \bar{\mathbf{u}}_{ni}}{\left(\bar{\mathbf{v}}_\infty + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} G_j \right) \cdot \bar{\mathbf{u}}_{ai}} \right] \quad (9.6.32)$$

Equation (9.6.31) can be written for N different control points, one associated with each of the N horseshoe vortices used to synthesize the lifting surface or system of lifting surfaces. This provides a system of N nonlinear equations relating the N unknown dimensionless vortex strengths, G_i , to known properties of the wing. At angles of attack below stall, this system of nonlinear equations surrenders quickly to Newton's method.

To apply Newton's method, the system of equations is written in the vector form

$$\mathbf{f}(\mathbf{G}) = \mathbf{R} \quad (9.6.33)$$

where

$$f_i(\mathbf{G}) = 2 \left| \bar{\mathbf{v}}_\infty + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} G_j \right| \times \bar{\boldsymbol{\zeta}}_i \left| G_i - \tilde{C}_{Li}(\alpha_i, \delta_i) \right| \quad (9.6.34)$$

and $\mathbf{R}$ is a vector of residuals. We wish to find the vector of dimensionless vortex strengths, $\mathbf{G}$, that will make all components of the residual vector, $\mathbf{R}$, go to zero. Thus, we want the change in the residual vector to be $-\mathbf{R}$. We start with an initial estimate for the $\mathbf{G}$ vector and iteratively refine the estimate by applying the Newton corrector equation

$$[\mathbf{J}] \Delta \mathbf{G} = -\mathbf{R} \quad (9.6.35)$$

where $[\mathbf{J}]$ is an N by N matrix of partial derivatives obtained by differentiating Eq. (9.6.34),

$$J_{ij} = \frac{\partial f_i}{\partial G_j} = \begin{cases} \frac{2 \bar{\mathbf{w}}_i \cdot (\bar{\mathbf{v}}_{ji} \times \bar{\boldsymbol{\zeta}}_i)}{|\bar{\mathbf{w}}_i|} G_i - \frac{\partial \tilde{C}_{Li}}{\partial \alpha_i} \frac{v_{ai} (\bar{\mathbf{v}}_{ji} \cdot \bar{\mathbf{u}}_{ni}) - v_{ni} (\bar{\mathbf{v}}_{ji} \cdot \bar{\mathbf{u}}_{ai})}{v_{ai}^2 + v_{ni}^2}, & j \neq i \\ 2 |\bar{\mathbf{w}}_i| + \frac{2 \bar{\mathbf{w}}_i \cdot (\bar{\mathbf{v}}_{ji} \times \bar{\boldsymbol{\zeta}}_i)}{|\bar{\mathbf{w}}_i|} G_i - \frac{\partial \tilde{C}_{Li}}{\partial \alpha_i} \frac{v_{ai} (\bar{\mathbf{v}}_{ji} \cdot \bar{\mathbf{u}}_{ni}) - v_{ni} (\bar{\mathbf{v}}_{ji} \cdot \bar{\mathbf{u}}_{ai})}{v_{ai}^2 + v_{ni}^2}, & j = i, \end{cases} \quad (9.6.36)$$

where

$$\bar{\mathbf{w}}_i \equiv \bar{\mathbf{v}}_i \times \bar{\boldsymbol{\zeta}}_i, \quad v_{ni} \equiv \bar{\mathbf{v}}_i \cdot \bar{\mathbf{u}}_{ni}, \quad v_{ai} \equiv \bar{\mathbf{v}}_i \cdot \bar{\mathbf{u}}_{ai}, \quad \bar{\mathbf{v}}_i \equiv \bar{\mathbf{v}}_\infty + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} G_j$$

Using Eq. (9.6.36) in Eq. (9.6.35), we compute the correction vector, $\Delta \mathbf{G}$. This correction vector is used to obtain an improved estimate for the dimensionless vortex strength vector, $\mathbf{G}$, according to the relation

$$\mathbf{G} = \mathbf{G} + \Omega \Delta \mathbf{G} \quad (9.6.37)$$

where Ω is a relaxation factor. The process is repeated until the magnitude of the largest residual is less than some convergence criteria. For angles of attack below stall, this method converges very rapidly using almost any initial estimate for the $\mathbf{G}$ vector with a relaxation factor of unity. At angles of attack beyond stall, the method must be highly under-relaxed and is very sensitive to the initial estimate for the $\mathbf{G}$ vector.

For the fastest convergence of Newton's method, we require an accurate initial estimate for the dimensionless vortex strength vector. For this purpose, a linearized version of Eq. (9.6.31) is useful. Furthermore, for wings of high aspect ratio at small angles of attack, the nonlinear terms in Eq. (9.6.31) are quite small and the linearized system can be used directly to give an accurate prediction of aerodynamic performance.

At small angles of attack, the airfoil section lift coefficient is very nearly a linear function of angle of attack. Thus, the relation for local section lift coefficient implied by Eq. (9.6.28) can be approximated as

$$\tilde{C}_{Li}(\alpha_i, \delta_i) \cong \tilde{C}_{Li,\alpha}(\alpha_i - \alpha_{L0i} + \varepsilon_i \delta_i) \quad (9.6.38)$$

where $\tilde{C}_{Li,\alpha}$, α_{L0i} , and ε_i are respectively the airfoil section lift slope, the zero lift angle of attack with no flap deflection, and the flap effectiveness, all evaluated for airfoil section i .

For wings of reasonably high aspect ratio at small angles of attack, the downwash is small and the local angle of attack can also be approximated as a linear function of the dimensionless vortex strengths. From Eq. (9.6.32), we have

$$\tan \alpha_i = \frac{v_{ni}}{v_{ai}}, \quad \cos \alpha_i = \frac{v_{ai}}{v_i}, \quad \sin \alpha_i = \frac{v_{ni}}{v_i},$$

where

$$v_{ni} \equiv \bar{\mathbf{v}}_i \cdot \bar{\mathbf{u}}_{ni}, \quad v_{ai} \equiv \bar{\mathbf{v}}_i \cdot \bar{\mathbf{u}}_{ai}, \quad v_i \equiv \sqrt{v_{ai}^2 + v_{ni}^2}, \quad \bar{\mathbf{v}}_i \equiv \bar{\mathbf{v}}_\infty + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} G_j$$

Using the small angle approximation for both the geometric angle of attack and the induced angle of attack, we can write

$$v_i \cong |\bar{\mathbf{v}}_\infty| = 1, \quad \cos \alpha_i \cong 1, \quad \alpha_i \cong \sin \alpha_i = \frac{v_{ni}}{v_i} \cong v_{ni} = \left(\bar{\mathbf{v}}_\infty + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} G_j \right) \cdot \bar{\mathbf{u}}_{ni}$$

Using this approximation in Eq. (9.6.38) to eliminate the local angle of attack, we have

$$\tilde{C}_{Li} \cong \tilde{C}_{Li,\alpha} \left(\bar{\mathbf{v}}_\infty \cdot \bar{\mathbf{u}}_{ni} + \sum_{j=1}^N \bar{\mathbf{v}}_{ji} \cdot \bar{\mathbf{u}}_{ni} G_j - \alpha_{L0i} + \varepsilon_i \delta_i \right) \quad (9.6.39)$$

Applying Eq. (9.6.39) to Eq. (9.6.31) and neglecting the remaining second order terms, we obtain the linearized system,

$$2|\bar{\mathbf{v}}_\infty \times \bar{\boldsymbol{\zeta}}_i| G_i - \tilde{C}_{Li,\alpha} \sum_{j=1}^N \bar{\mathbf{v}}_{ji} \cdot \bar{\mathbf{u}}_{ni} G_j = \tilde{C}_{Li,\alpha} (\bar{\mathbf{v}}_\infty \cdot \bar{\mathbf{u}}_{ni} - \alpha_{L0i} + \varepsilon_i \delta_i) \quad (9.6.40)$$

Equation (9.6.40) gives good results, at small angles of attack, for wings of reasonably high aspect ratio and little sweep. For larger angles of attack or highly swept wings, the nonlinear system of equations given by Eq. (9.6.31) should be used. However, at angles of attack below stall, Eq. (9.6.40) still provides a reasonable initial estimate for the dimensionless vortex strength vector, to be used with Newton's method for obtaining a solution to the nonlinear system.

Once the vortex strengths have been determined from either Eq. (9.6.31) or Eq. (9.6.40), the total aerodynamic force vector can be determined from Eq. (9.6.27). If the lifting surface or surfaces are synthesized from a large number of horseshoe vortices, each covering a small spanwise increment of one lifting surface, we can approximate the aerodynamic force as being constant over each spanwise increment. Then, from Eq. (9.6.27), the total aerodynamic force is given by

$$\bar{\mathbf{F}} = \rho \sum_{i=1}^N \left(\Gamma_i \bar{\mathbf{v}}_\infty + \sum_{j=1}^N \frac{\Gamma_i \Gamma_j}{\bar{c}_j} \bar{\mathbf{v}}_{ji} \right) \times \delta \bar{\mathbf{l}}_i \quad (9.6.41)$$

where $\delta \bar{\mathbf{l}}_i$ is the spatial vector along the bound segment of horseshoe vortex i from node 1 to node 2.

Nondimensionalizing Eq. (9.6.41), the **total nondimensional aerodynamic force vector** is

$$\frac{\bar{\mathbf{F}}}{\frac{1}{2} \rho V_\infty^2 A_r} = 2 \sum_{i=1}^N \left(G_i \bar{\mathbf{v}}_\infty + \sum_{j=1}^N G_i G_j \bar{\mathbf{v}}_{ji} \right) \times \bar{\boldsymbol{\zeta}}_i \frac{\delta S_i}{S_r} \quad (9.6.42)$$

where S_r is the global reference area, δS_i is the planform area of the spanwise increment of the lifting surface covered by horseshoe vortex i

$$\delta S_i = \int_{s=s_1}^{s_2} c \, ds \quad (9.6.43)$$

c is the local chord length, and s is the spanwise coordinate.

The aerodynamic moment generated about the center of gravity is

$$\bar{\mathbf{M}} = \rho \sum_{i=1}^N \bar{\mathbf{r}}_i \times \left[\left(\Gamma_i \bar{\mathbf{v}}_\infty + \sum_{j=1}^N \frac{\Gamma_i \Gamma_j}{\bar{c}_j} \bar{\mathbf{v}}_{ji} \right) \times \bar{\boldsymbol{\zeta}}_i \right] + \delta \bar{\mathbf{M}}_i \quad (9.6.44)$$

where $\bar{\mathbf{r}}_i$ is the spatial vector from the center of gravity to control point i and $\delta \bar{\mathbf{M}}_i$ is the quarter-chord moment generated by the spanwise increment of the wing covered by horseshoe vortex i . If we assume a constant section moment coefficient over each spanwise increment, then

$$\delta \bar{\mathbf{M}}_i = -\frac{1}{2} \rho V_\infty^2 \tilde{C}_{mi} \int_{s=s_1}^{s_2} c^2 \, ds \, \bar{\mathbf{u}}_{si} \quad (9.6.45)$$

where $\tilde{C}_{mi}$ is the local section moment coefficient, $\bar{\mathbf{u}}_{si}$ is the local unit vector in the spanwise direction as shown in Fig. 9.6.6 and defined by

$$\bar{\mathbf{u}}_{si} = \bar{\mathbf{u}}_{ai} \times \bar{\mathbf{u}}_{ni} \quad (9.6.46)$$

Using Eq. (9.6.45) in Eq. (9.6.44) and nondimensionalizing gives

$$\frac{\bar{\mathbf{M}}}{\frac{1}{2} \rho V_\infty^2 S_r l_r} = \sum_{i=1}^N \left\{ 2 \bar{\mathbf{r}}_i \times \left[\left(G_i \bar{\mathbf{v}}_\infty + \sum_{j=1}^N G_i G_j \bar{\mathbf{v}}_{ji} \right) \times \bar{\boldsymbol{\zeta}}_i \right] - \frac{\tilde{C}_{mi}}{\delta S_i} \int_{s=s_1}^{s_2} c^2 \, ds \, \bar{\mathbf{u}}_{si} \right\} \frac{\delta S_i}{S_r l_r} \quad (9.6.47)$$

where l_r is the global reference length.

To this point, the local characteristic length, $\bar{c}_i$, has not been defined. It could be any characteristic length associated with the spanwise increment of the wing covered by horseshoe vortex i . From Eq. (9.6.47), the most natural choice for this local characteristic length is the integral of the chord length squared, with respect to the spanwise coordinate, divided by the incremental area. If we also assume a linear variation in chord length over each spanwise increment, we then have

$$\delta S_i = \int_{s=s_1}^{s_2} c \, ds = \frac{c_{i_1} + c_{i_2}}{2} (s_{i_2} - s_{i_1}) \quad (9.6.48)$$

$$\bar{c}_i = \frac{1}{\delta S_i} \int_{s=s_1}^{s_2} c^2 \, ds = \frac{2}{3} \frac{c_{i_1}^2 + c_{i_1} c_{i_2} + c_{i_2}^2}{c_{i_1} + c_{i_2}} \quad (9.6.49)$$

With these definitions, the **dimensionless aerodynamic moment generated about the center of gravity** is

$$\frac{\bar{\mathbf{M}}}{\frac{1}{2} \rho V_\infty^2 S_r l_r} = \sum_{i=1}^N \left\{ 2 \bar{\mathbf{r}}_i \times \left[\left(G_i \bar{\mathbf{v}}_\infty + \sum_{j=1}^N G_i G_j \bar{\mathbf{v}}_{ji} \right) \times \bar{\boldsymbol{\zeta}}_i \right] - \tilde{C}_{mi} \bar{c}_i \bar{\mathbf{u}}_{si} \right\} \frac{\delta S_i}{S_r l_r} \quad (9.6.50)$$

Once the dimensionless vortex strengths, G_i , are known, Eqs. (9.6.42) and (9.6.50) are used to evaluate the aerodynamic forces and moments.

Each lifting surface must, of course, be divided into spanwise elements, in a manner similar to that shown symbolically in Fig. 9.6.4. In this figure, the wing is divided into elements of equal spanwise increment. However, this is not the most efficient way in which to grid a wing. Since vorticity is shed from a wing more rapidly in the region near the wingtips, for best computational efficiency, the nodal points should be clustered more tightly in this region. Conventional cosine clustering has been found to be quite efficient for this purpose. For straight wings, clustering is only needed near the wingtips and the cosine distribution can be applied across the entire span of the wing. However, for wings with sweep or dihedral, there is a step change in the slope of the quarter-chord at the root of the wing. This step change causes the downwash to change very rapidly in the region near the root. Thus, in general, it is recommended that cosine clustering be applied independently over each semi-span of a wing. This clusters the nodes more tightly at both the tip and at the root. This clustering is based on the change of variables,

$$\frac{s}{b} = \frac{[1 - \cos(\theta)]}{4} \quad (9.6.51)$$

where s is the spanwise coordinate and b is twice the semi-span. Over each semi-span, θ varies from zero to π as s varies from zero to $b/2$. Distributing the nodes uniformly in θ will provide the desired clustering in s . If the total number of horseshoe elements desired on each semi-span is n , the spanwise nodal coordinates are computed from,

$$\frac{s_i}{b} = \frac{1}{4} \left[1 - \cos\left(\frac{i\pi}{n}\right) \right], \quad 0 \leq i \leq n \quad (9.6.52)$$

where the bound segment of horseshoe vortex i extends from node i to node $i-1$ on any left semi-span and from node $i-1$ to node i on any right semi-span. Using this nodal distribution with about 40 horseshoe elements per semi-span gives the best compromise between speed and accuracy. Figure 9.6.7 shows a system of lifting surfaces overlaid with a grid of this type using 20 elements per semi-span.

For maximum accuracy and computational efficiency, some attention must also be paid to the location of control points. At first thought, it would seem most reasonable to place control points on the bound segment of each vortex, midway between the two trailing legs. However, it has been found that this does not give the best accuracy. A significant improvement in accuracy, for a given number of elements, can be achieved by placing the control points midway in θ rather than midway in s . Thus, the spanwise control point coordinates should be computed from,

$$\frac{s_i}{b} = \frac{1}{4} \left[1 - \cos\left(\frac{i\pi}{n} - \frac{\pi}{2n}\right) \right], \quad 1 \leq i \leq n \quad (9.6.53)$$

This distribution places control points very near the spatial midpoint of each bound vortex segment, over most of the wing. However, near the root and the tip, these control points are significant offset from the spatial midpoint.

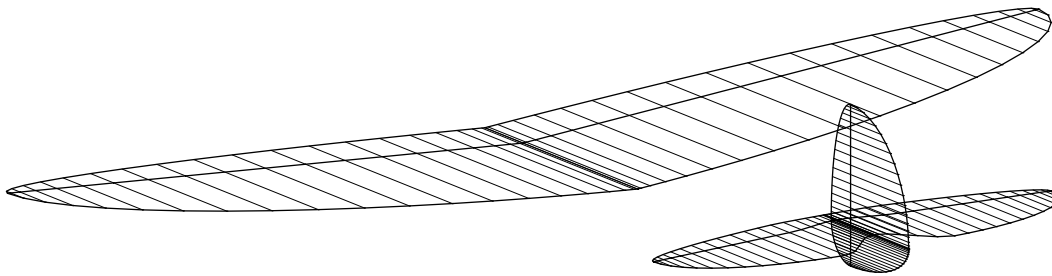


Figure 9.6.7. Lifting-line grid with cosine clustering and 20 elements per semi-span.

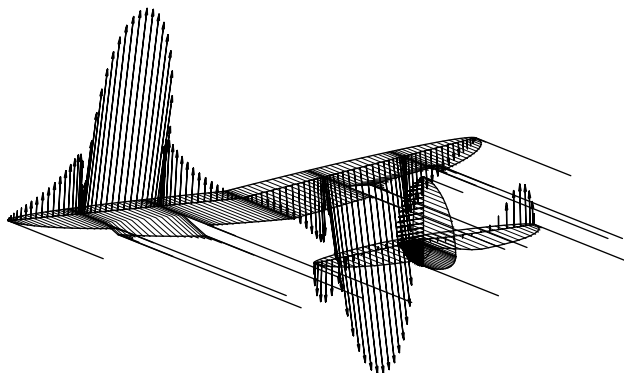


Figure 9.6.8. Pressure forces and shed vorticity for a system of lifting surfaces as predicted by the numerical lifting-line method.

This numerical lifting-line method can be used to predict the aerodynamic forces and moments acting on a system of lifting surfaces with arbitrary position and orientation. Each lifting surface is synthesized by distributing horseshoe vortices along the quarter-chord in the manner shown in Fig. 9.6.7. Because all of the horseshoe vortices used to synthesize the complete system of lifting surfaces are combined and forced to satisfy either Eq. (9.6.31) or Eq. (9.6.40) as a single system of coupled equations, the interactions between lifting surfaces are directly accounted for. An example of such interactions can be seen in Fig. 9.6.8. The lifting surfaces shown in this figure all have symmetric airfoil sections and are oriented to give a zero geometric angle of attack. The only direct production of lift in this configuration comes from a deflection of the ailerons. The deflected ailerons produce lift and vorticity that in turn induce lift and vorticity on all other surfaces. The arrows in Fig. 9.6.8 indicate the magnitude and direction of the local section force and the stream-wise lines at the trailing edge of each lifting surface indicate the magnitude of the local shed vorticity.

As we have previously seen, for straight elliptic wings, Prandtl's classical lifting-line theory gives a very simple closed form solution. As a first order test, the numerical lifting-line method should agree with this closed form solution, at small angles of attack, for an elliptic wing with no sweep, no dihedral, and no geometric or aerodynamic twist. Figures 9.6.9 and 9.6.10 show a comparison between the section lift distribution and the section induced drag distribution predicted by Prandtl's closed form solution and that predicted by the numerical lifting-line method, for three straight elliptic wings of different aspect ratio. The agreement in all cases is excellent.

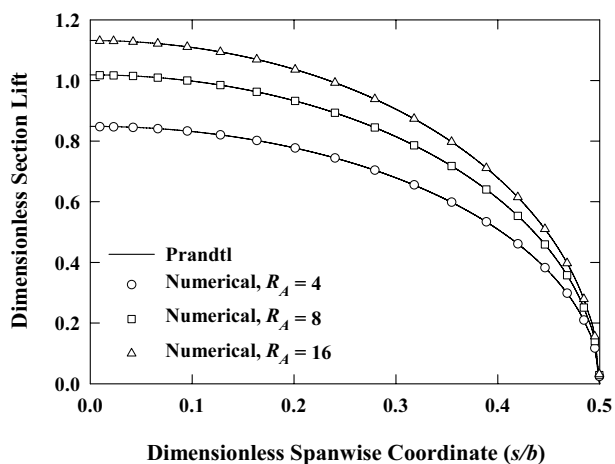


Figure 9.6.9. The section lift distribution for three elliptic wings.

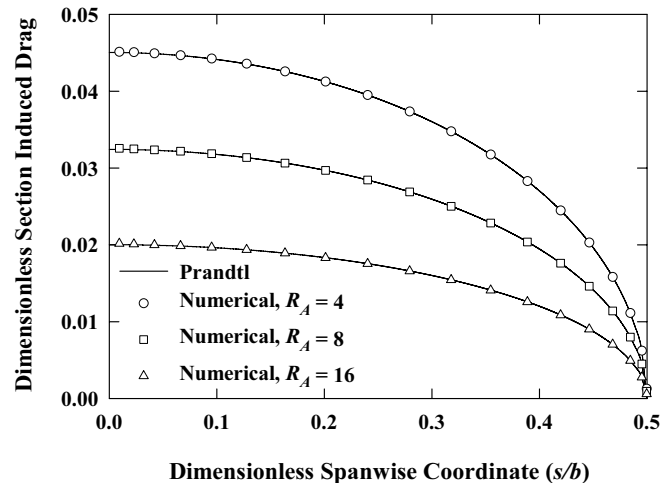


Figure 9.6.10. The section induced drag distribution for three elliptic wings.

Unlike the closed form solution to Prandtl's classical lifting-line theory, the numerical lifting-line method can be applied to wings with sweep and/or dihedral. To examine how well the numerical lifting-line method predicts the effects of sweep, Phillips and Snyder (2000) compared results obtained from this method, a numerical panel method, and an inviscid CFD solution. These results were also compared with experimental data for two different wings. This comparison is shown in Fig. 9.6.11. The solid line and filled symbols correspond to a straight wing of aspect ratio 6.57, with experimental data obtained from McAlister and Takahashi (1991). The dashed line and open symbols are for a 45-degree swept wing of aspect ratio 5.0, having experimental data reported by Weber and Brebner (1958). Both wings have symmetric airfoil sections and constant chord throughout the span, with no geometric twist. The straight wing has a thickness of 15 percent and the swept wing has a thickness of 12 percent. From the results shown in Fig. 9.6.11, we see that the lift coefficient predicted by all four methods is in good agreement with experimental observations for both wings. However, the computational time required to obtain a solution using the numerical panel method was about 2.5×10^4 times that required for the lifting-line method and the inviscid CFD solutions required approximately 2.7×10^6 times as long as the lift-line solutions. The accuracy of the numerical lifting line method for predicting dihedral effects was also investigated by Phillips and Snyder (2000) and the results obtained were similar to those shown in Fig. 9.6.11.

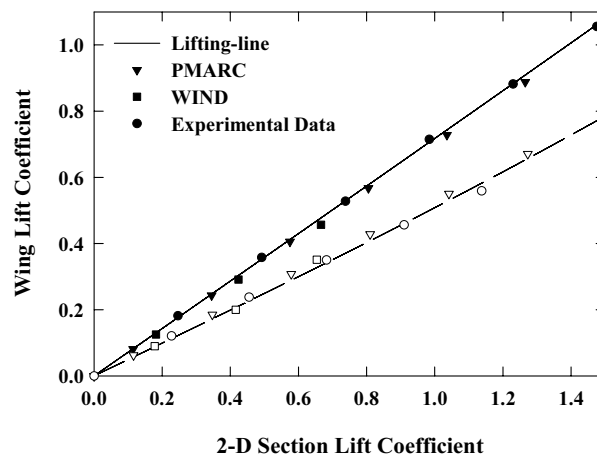


Figure 9.6.11. Comparison between the lift coefficient predicted by the numerical lifting-line method, a numerical panel method (PMARC), and an inviscid CFD solutions (WIND) with that obtained from experimental data, for a straight wing and a wing with 45 degrees of sweep.

The insight of Ludwig Prandtl (1875-1953) was nothing short of astonishing. This was never more dramatically demonstrated than in the development of his classical lifting-line theory, during the period 1911 through 1918. The utility of this simple and elegant theory is so great that it is still widely used today. Furthermore, with a few minor alterations and the use of a modern computer, the model proposed by Prandtl can be used to predict the inviscid forces and moments acting on lifting surfaces of aspect ratio greater than about four with an accuracy as good as that obtained from modern panel codes or CFD, but at a small fraction of the computational cost.

Like panel methods, lifting-line theory provides only a potential flow solution. Thus, the forces and moments computed from this method do not include viscous effects, so the parasitic drag is unobtainable. In addition to this restriction, which also applies to panel methods, lifting-line theory imposes an additional restriction, which does not apply to panel methods. For lifting surfaces with low aspect ratio, Prandtl's hypothesis breaks down and the usual relationship between local section lift and local section angle of attack no longer applies. It has long been established that lifting-line theory gives good agreement with experimental data for lifting surfaces of aspect ratio greater than about four. For lifting surfaces of lower aspect ratio panel methods or CFD solutions should be used.

The numerical lifting-line method contains no inherent requirement for a linear relationship between section lift and section angle of attack. Thus, the method can be applied, with caution, to account approximately for the effects of stall. The lifting-line method requires a known relationship for the section lift coefficient as a function of section angle of attack. Since such relationships are obtained experimentally beyond stall, the numerical lifting-line method predicts stall by using a semi-empirical correction to an otherwise potential flow solution. For this reason, the method should be used with extreme caution for angles of attack beyond stall. However, the method does provide a reasonable means for predicting the onset of stall.